

Model Question Paper-I with effect from 2025-26 (CBCS Scheme)

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USN

Fourth Semester B.E. Degree Examination Database Management System

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			CO	BL	Marks
Q. 01	a	What is a Database? Explain the characteristics of the Database approach.	CO1	L2	8
	b	Explain the three-schema architecture with a neat diagram.	CO1	L2	6
	c	Explain the difference between the two-tier and three-tier client/server architectures?	CO1	L2	6
OR					
Q. 02	a	Explain the component modules of DBMS & their interactions with diagrams.	CO1	L2	8
	b	Draw and explain ER modeling concepts are used to represent a Company database schema. Describe the role of entities, attributes, primary keys, and structural constraints such as cardinality and participation ratios with suitable examples.	CO1	L2	6
	c	With suitable examples, explain different types of attributes.	CO1	L2	6
Module-2					
Q. 03	a	Consider the following relational database schema and write the queries in relational algebra expressions: EMP(Eno, Ename, Salary, Address, Phone, DNo) DEPT(DNo, Dname, DLoc, MgrEno) DEPENDENT(Eno, Dep_Name, Drelation, Dage) i) List all the employees who reside in 'Davangere'. ii) List all the employees who earn a salary between 30000 & 40000. iii) List all the employees who work for the 'Research' department. iv) List all the department names along with the names of the managers.	CO2	L3	8
	b	Explain the different types of update operations on relational database. How basic operation deals with constraint violation.	CO2	L2	6
	c	Explain the Join operations with respect to binary relational algebra.	CO2	L2	6
OR					
Q. 04	a	Demonstrate the ER-to-Relational mapping algorithm by converting an ER diagram into relational tables, showing each step with examples.	CO2	L3	8
	b	Explain Unary relational operations with examples.	CO2	L2	6
	c	Illustrate the relational algebra operators with examples for select and project operation.	CO2	L2	6
Module-3					
	a	Explain the different SQL data types with suitable examples. Discuss how	CO3	L2	6

Q. 05		choosing appropriate data types affects database design.			
	b	Describe assertions and triggers in SQL . Highlight their differences and use cases with examples.	CO3	L2	6
	c	Discuss the working of SELECT statements in SQL . Explain the role of: - WHERE - GROUP BY - HAVING - ORDER BY with examples.	CO3	L3	8
OR					
Q. 06	a	Explain the concept of views in SQL . Discuss: - Advantages - Types of views - How views provide security	CO3	L2	6
	b	Explain different types of SQL joins with examples: - Inner join - Left outer join - Right outer join - Full outer join	CO3	L2	6
	c	Given: Employee(EID, Name, Salary, DeptID) Write SQL queries to: - Increase salary by 10% for all employees in a specific department - Delete employees with salary less than 3000 - Insert a new employee record - Retrieve employees earning more than average salary	CO3	L3	8
Module-4					
Q. 07	a	Define Join Dependency and Fifth Normal Form (5NF). Illustrate with an example why a relation in 4NF may not be in 5NF, and show the lossless decomposition.	CO3	L2	6
	b	Discuss the desirable properties of transactions (ACID properties). Elaborate on each property with real-world database scenarios.	CO4	L3	7
	c	Given relation R(A,B,C,D,E) with F = {A→BC, CD→E, B→D, E→A}. Compute all non-trivial candidate keys using attribute closures. Prove that {A,E} is a superkey but not a candidate key and find a minimal cover.	CO3	L3	7
OR					
Q. 08	a	What is Boyce-Codd Normal Form (BCNF)? Differentiate it from 3NF with a suitable example where a relation is in 3NF but not in BCNF.	CO3	L2	6
	b	What is transaction processing? Explain the key transaction and system concepts, such as recovery and concurrency control.	CO4	L3	7
	c	Describe transaction support in SQL, including commands like COMMIT, ROLLBACK and SAVEPOINT. Provide syntax and an example of nested transactions.	CO4	L3	7
Module-5					
	a	Explain the concept of the Two-Phase Locking (2PL) protocol used in database concurrency control. Discuss how this protocol ensures conflict serializability in transaction scheduling.	CO4	L2	8

Q. 09	b	Describe the key characteristics of NoSQL databases and explain how they differ from traditional relational database systems.	CO5	L2	6
	c	Explain the primary CRUD (Create, Read, Update, Delete) operations in MongoDB with suitable examples.	CO5	L2	6
OR					
Q. 10	a	Explain the concepts of deadlock and starvation in database systems. Discuss the various approaches used to prevent, avoid, detect and recover from these problems in concurrency control.	CO4	L2	8
	b	Describe the CAP theorem in distributed database systems. Which of the three properties are most important in NOSQL systems?	CO5	L2	6
	c	Explain the working of the Voldemort key–value distributed data store. Illustrate its data storage and retrieval mechanism with a suitable example.	CO5	L2	6